

Osmosis and Osmotic pressure of the solution

41. Blood cells will remain as such in

- (a) Hypertonic solution
- (b) Hypotonic solution
- (c) Isotonic solution
- (d) None of these

42. The osmotic pressure of a dilute solution is directly proportional to the

- (a) Diffusion rate of the solute
- (b) Ionic concentration
- (c) Elevation of B.P.
- (d) Flow of solvent from a concentrated to a dilute solution

43. The osmotic pressure in atmospheres of 10% solution of canesugar at 69°C is

- (a) 724
- (b) 824
- (c) 8.21
- (d) 7.21

44. Which of the following molecules would diffuse through a cell membrane

- (a) Fructose
- (b) Glycogen
- (c) Haemoglobin
- (d) Catalase

45. Two solutions of KNO_3 and CH_3COOH are prepared separately. Molarity of both is 0.1M and osmotic pressures are P_1 and P_2 respectively.

The correct relationship between the osmotic pressures is

- (a) $P_2 > P_1$
- (b) $P_1 = P_2$
- (c) $P_1 > P_2$
- (d) $\frac{P_1}{P_1+P_2} = \frac{P_2}{P_1+P_2}$

46. The osmotic pressure of a dilute solution of a non-volatile solute is

- (a) Directly proportional to its temperature on the centigrade scale
- (b) Inversely proportional to its temperature on the Kelvin scale
- (c) Directly proportional to its temperature on the Kelvin scale
- (d) Inversely proportional to its temperature on the centigrade scale

47. Osmotic pressure of a urea solution at 10°C is 500mm . Osmotic pressure of the solution become 105.3 mm . When it is diluted and temperature raised to 25°C . The extent of dilution is

- (a) 6 Times
- (b) 5 Times
- (c) 7 Times
- (d) 4 Times

48. If a 0.1M solution of glucose (mol. wt. 180) and 0.1molar solution of urea (mol. wt. 60) are placed on the two



- sides of a semipermeable membrane to equal heights, then it will be correct to say
- There will be no net movement across the membrane
 - Glucose will flow across the membrane into urea solution
 - Urea will flow across the membrane into glucose solution
 - Water will flow from urea solution into glucose solution
49. At constant temperature, the osmotic pressure of a solution
- Directly proportional to the concentration
 - Inversely proportional to the concentration
 - Directly proportional to the square of the concentration
 - Directly proportional to the square root of the concentration
50. The solution containing 4.0 gm of a polyvinyl chloride polymer in 1 litre of dioxane was found to have an osmotic pressure 6.0×10^{-4} atmosphere at 300K, the value of R used is 0.082 litre atmosphere $\text{mole}^{-1}\text{K}^{-1}$. The molecular mass of the polymer was found to be
- 3.0×10^2
 - 1.6×10^5
 - 5.6×10^4
 - 6.4×10^2
51. Solvent molecules pass through the semipermeable membrane is called
- Electrolysis
 - Electrophoresis
 - Cataphoresis
 - Osmosis
52. If molecular weight of compound is increased then sensitivity is decreased in which of the following methods
- Elevation in boiling point
 - Viscosity
 - Osmosis
 - Dialysis
53. If solubility of NaCl at 20°C is 35 gm per 100 gm of water. Then on adding 50 gm of NaCl to the same volume at same temperature the salt remains undissolved is
- 15 gm
 - 20 gm
 - 50 gm
 - 35 gm
54. Which of the following associated with isotonic solution is not correct
- They will have the same osmotic pressure
 - They have the same weight concentration
 - Osmosis does not take place when the two solutions are



- separated by a semipermeable membrane
- (d) They will have the same vapour pressure
55. If osmotic pressure of a solution is 2 atm at 273 K , then at 546 K , the osmotic pressure is
- (a) 0.5 atm (b) 1 atm
(c) 2 atm (d) 4 atm
56. In osmosis reaction, the volume of solution
- (a) Decreases slowly
(b) Increases slowly
(c) Suddenly increases
(d) No change
57. As a result of osmosis the volume of solution
- (a) Increases
(c) Decreases
(c) Remains constant
(d) Increases or decreases
58. A solution of urea contain 8.6 gm/litre ($\text{mol. wt. } 60.0$). It is isotonic with a 5% solution of a non-volatile solute. The molecular weight of the solute will be
- (a) 348.9 (b) 34.89
(c) 3489 (d) 861.2
59. One mole each of urea, glucose and sodium chloride were dissolved in one litre of water. Equal osmotic pressure will be produced by solutions of
- (a) Glucose and sodium chloride
(b) Urea and glucose
(c) Sodium chloride and urea
(d) None of these
60. Which of the following aqueous solutions produce the same osmotic pressure
- (a) 0.1 M NaCl solution
(b) 0.1 M glucose solution
(c) 0.6 g urea in 100 ml solution
(d) 1.0 g of a non-electrolyte solute (X) in 50 ml solution (Molar mass of $X = 200$)
61. Which of the following aqueous solutions are isotonic ($R = 0.082\text{ atm}$)
- (a) 0.01 M glucose
(b) 0.01 M NaNO_3
(c) 500 ml solution containing 0.3 g urea
(d) 0.04 NHCl

